

Pemphigus Vulgaris — Study Notes (Q69–Q72)

Dermatology | Undergraduate Medical Reference

69. Why does pemphigus vulgaris (PV) commonly present first with oral lesions?

- Desmoglein 3 (Dsg3) is the dominant desmosomal adhesion protein in oral mucosa, while skin expresses both Dsg1 and Dsg3.
- The "desmoglein compensation theory" explains this: in skin, Dsg1 can compensate for loss of Dsg3 function, so skin stays intact early on. In oral mucosa, Dsg1 expression is low/absent, so anti-Dsg3 antibodies alone are enough to cause acantholysis.
- Hence, patients typically present first with painful oral erosions (often the presenting and sometimes only complaint for weeks to months) before skin lesions appear, since early PV is usually anti-Dsg3-antibody dominant.

Desmoglein Compensation Theory: Why Oral Lesions Appear First in PV

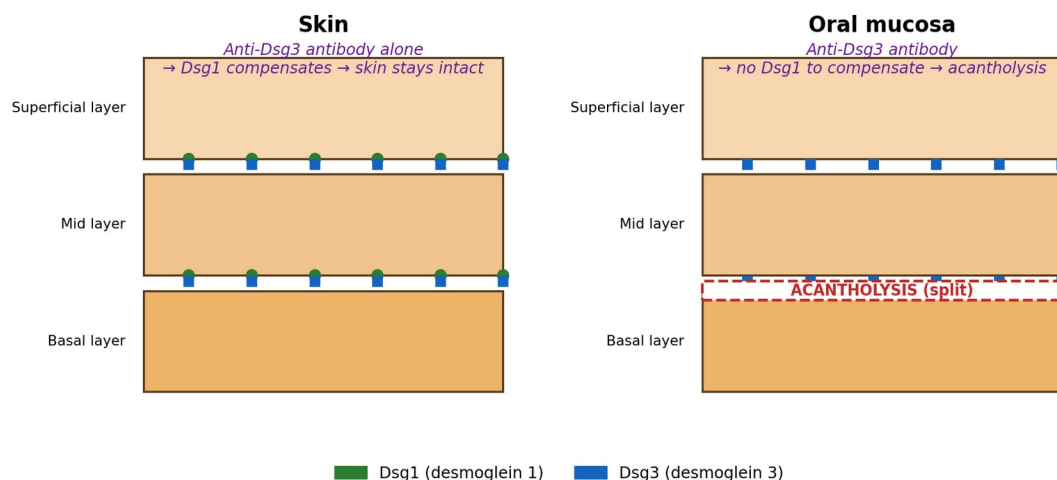


Figure 1. Desmoglein compensation theory — schematic showing why anti-Dsg3 antibodies alone cause acantholysis in mucosa but not skin.

70. What is the classic appearance of pemphigus vulgaris skin lesions?

- Flaccid, thin-walled bullae on normal-appearing or erythematous skin — they rupture very easily (acantholysis is suprabasal, so the blister roof is only the upper epidermis).
- Because they rupture quickly, patients often present with painful erosions and crusting rather than intact bullae.
- **Nikolsky sign positive** and **Asboe-Hansen sign positive** (see table below).
- Lesions favor scalp, face, chest, axillae, groin — areas of trauma/friction.
- Heal without scarring (unless secondarily infected) since the split is intraepidermal, not at the basement membrane.

Sign	How it's elicited	Significance
Nikolsky sign	Lateral shearing/rubbing pressure applied to normal-appearing perilesional skin	Positive = outer epidermis shears off, exposing a moist base → indicates fragile intraepidermal adhesion
Asboe-Hansen sign (bulla spread sign)	Firm pressure applied on top of an intact, existing bulla	Positive = fluid dissects laterally, causing the blister to enlarge/spread beyond the point of pressure

Site of Blister Formation in Pemphigus Vulgaris (Suprabasal Acantholysis)

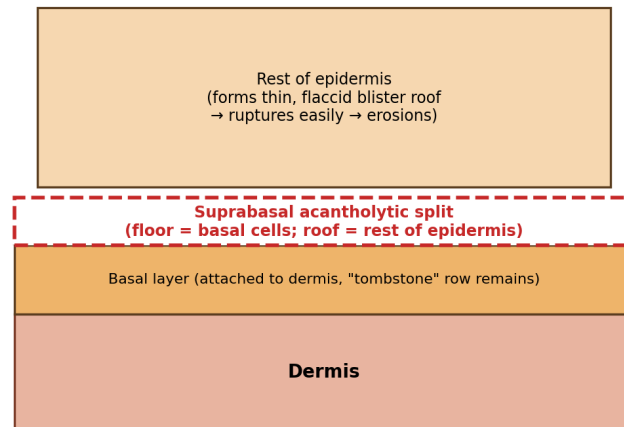


Figure 2. Site of blister formation in PV — suprabasal acantholytic split producing a thin, fragile blister roof.

71. What is pemphigus vegetans?

- An uncommon, milder clinical variant of PV.
- Instead of flaccid bullae/erosions, lesions evolve into hypertrophic, vegetating, verrucous plaques with pustules, typically in intertriginous/flexural areas (axillae, groin, inframammary folds, umbilicus) and around body orifices.

Two recognized subtypes:

- **Neumann type** – begins like typical PV with vesicles/bullae that later develop into vegetating plaques.
- **Hallopeau type** – begins with pustules (not bullae) that evolve into vegetations; generally has a more benign course.
- Histology shows acantholysis plus marked epidermal hyperplasia (pseudoepitheliomatous hyperplasia) with eosinophilic microabscesses.
- Immunofluorescence findings are the same as PV (intercellular IgG/C3 deposits, anti-Dsg3 ± anti-Dsg1 antibodies).

72. What is paraneoplastic pemphigus and how does it differ clinically?

An autoimmune bullous disease occurring in association with an underlying neoplasm, most commonly non-Hodgkin lymphoma, chronic lymphocytic leukemia (CLL), Castleman disease, thymoma, and less commonly other malignancies.

Antibodies target multiple antigens — not just desmogleins but also plakin family proteins (desmoplakin I/II, envoplakin, periplakin, BP230, plectin) in addition to Dsg3 and Dsg1.

How it differs clinically from PV:

- Oral/mucosal involvement is more severe and extensive — painful, treatment-resistant stomatitis involving the entire oropharynx, often with a hallmark severe, painful hemorrhagic cheilitis — more intense than typical PV.
- Polymorphous skin lesions — can mimic erythema multiforme, lichen planus, graft-versus-host disease, or bullous pemphigoid, rather than pure flaccid bullae.
- Frequent conjunctival and other mucous membrane involvement (esophagus, genital mucosa), which can lead to scarring — unlike PV, which usually heals without scarring.
- Can involve lungs — bronchiolitis obliterans, which is a major cause of mortality and not seen in classic PV.
- Immunologically: shows both intercellular staining (like PV) AND antibodies against plakins detectable by immunoprecipitation/immunoblotting — this combination is diagnostic and distinguishes it serologically from PV.
- Prognosis is significantly worse, driven both by the underlying malignancy and by the disease itself (especially pulmonary involvement).
- Treatment often has a poorer response to standard immunosuppression compared to PV, and management must address the associated neoplasm.

Summary Comparison Table

Feature	Pemphigus Vulgaris	Pemphigus Vegetans	Paraneoplastic Pemphigus
Onset site	Oral mucosa first, then skin	Intertriginous/flexural skin	Severe oral + skin, often first sign of occult tumor
Primary lesion	Flaccid bullae → erosions	Vegetating, verrucous plaques with pustules	Polymorphous: bullae, erosions, EM/LP-like, GVHD-like
Oral involvement	Common, painful erosions	Less prominent	Severe, extensive; hallmark hemorrhagic cheilitis
Associated condition	None (primary autoimmune)	None (variant of PV)	Lymphoma, CLL, Castleman disease, thymoma
Antigens targeted	Dsg3 (± Dsg1)	Dsg3 (± Dsg1)	Dsg3, Dsg1 + plakins (desmoplakin, envoplakin, periplakin, BP230)
Histology	Suprabasal acantholysis	Acantholysis + pseudoepitheliomatous hyperplasia, eosinophilic microabscesses	Acantholysis + interface dermatitis/keratinocyte necrosis
Other organ	None typically	None typically	Lungs – bronchiolitis

Feature	Pemphigus Vulgaris	Pemphigus Vegetans	Paraneoplastic Pemphigus
involvement			obliterans (major mortality cause)
Scarring	Usually none	Usually none	Can scar (conjunctiva, mucosa)
Prognosis	Good with treatment	Generally better than PV	Poor; depends on tumor + pulmonary disease